

Cognitive Science — Module 08: Planning — Study Questions

1. What does "planning as inference" mean? How is planning similar to perception?
2. How does an agent evaluate different possible policies (action sequences)?
3. What is Expected Free Energy's role in planning? How does it help select the best policy?
4. What is a deep temporal model? Give an example of thinking at three different timescales simultaneously.
5. How do hierarchical models allow agents to plan at different timescales?
6. What is the relationship between imagination and perception in Active Inference?
7. Why do dreams feel real? How does Active Inference explain this?
8. What is counterfactual thinking? Give an example and explain why it's useful.
9. Define a habit in Active Inference terms. How does it differ from a deliberate plan?
10. Why are habits hard to break? What needs to happen in terms of the generative model?
11. What is the horizon problem in planning? Why can't agents plan infinitely far ahead?
12. Explain bounded rationality. How do agents deal with limited planning capacity?
13. How does pruning (considering only plausible policies) make planning feasible?
14. What is satisficing? How does it differ from optimizing?
15. How does hierarchical decomposition (breaking plans into subgoals) help with planning?
16. Compare planning in a chess game to planning your career. How do the timescales and strategies differ?
17. Can animals plan? Give evidence for and against planning in non-human species.
18. How does procrastination relate to planning and Expected Free Energy?
19. What is the relationship between memory and planning? Does imagining the future use the same neural systems as remembering the past?
20. Looking back at all 8 modules (Systems → Agents → Perception → Cognition → Action → Learning → Communication → Planning), how does each build on the previous one? Describe the logical chain.